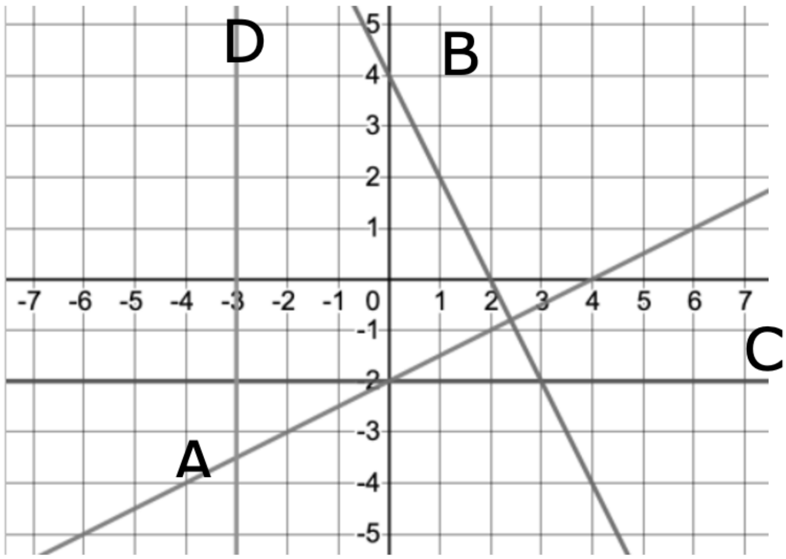
Grade 8
Unit 7
Lesson 8 Practice

Name___Answer key______
Date___________________
Period__________________

Use the following graph to answer problems 1–6:



1. Write the equation of line A in slope-intercept form.

   $y = \frac{1}{2}x - 2$

2. Write the equation of line B in slope-intercept form.

   $y = -2x + 4$

3. Which line is formed from the equation $2x - 4y = 8$?

   $2x - 4y = 8$

   $\underline{-2x} \qquad \underline{-2x}$

   $\frac{-4y}{-4} = \frac{-2x}{-4}\ \frac{+8}{-4}$

   $y = \frac{1}{2}x - 2$

   Line A

4. Write the equation of line C in slope-intercept form.

   $y = -2$

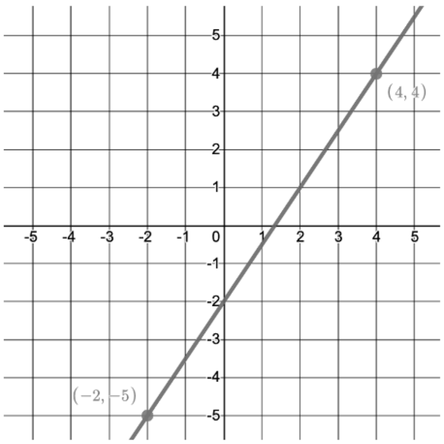
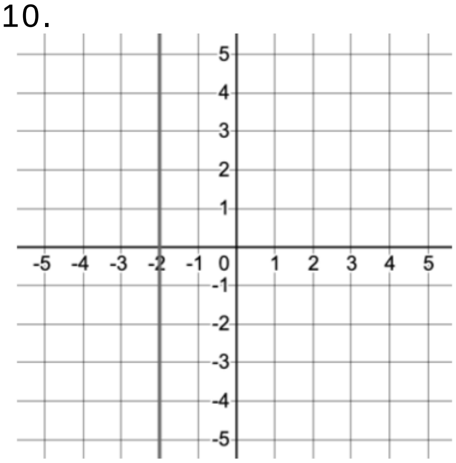
5. Explain why the equation of line D has no y-intercept.

   It does not cross the y-axis.

6. Write the equation of line D.

   $x = -3$

Find the slope and y-intercept of each line. Then, write the equation of each line in slope-intercept form:

<table>
<tr><th>Linear Relationship</th><th>Slope</th><th>y-intercept</th><th>Equation</th></tr>
<tr><td>7.

</td><td>$(-2,-5)$ $(4,4)$
$\frac{4-(-5)}{4-(-2)} = \frac{9}{6} = \frac{3}{2}$</td><td>$(0,-2)$</td><td>$y = \frac{3}{2}x - 2$</td></tr>
<tr><td>8.
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-8</td><td>-2</td></tr>
<tr><td>-4</td><td>-2</td></tr>
<tr><td>-2</td><td>-2</td></tr>
<tr><td>1</td><td>-2</td></tr>
</table></td><td>$(-8,-2)$ $(-4,-2)$
$\frac{-2-(-2)}{-4-(-8)} = \frac{0}{4}$
m = 0</td><td>$-2 = 1(0) + b$
$-2 = b$</td><td>$y = -2$</td></tr>
<tr><td>9.
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>5</td><td>-2</td></tr>
<tr><td>11</td><td>-6</td></tr>
<tr><td>14</td><td>-8</td></tr>
<tr><td>17</td><td>-10</td></tr>
</table></td><td>$(5,-2)$ $(11,-6)$
$\frac{-6-(-2)}{11-5} = \frac{-4}{6}$
$= \frac{-2}{3}$</td><td>$-2 = 5\left(\frac{-2}{3}\right) + b$
$-2 = \frac{-10}{3} + b$
$\frac{+10}{3}$ $\frac{+10}{3}$
$\frac{4}{3} = b$</td><td>$y = \frac{-2}{3}x + \frac{4}{3}$</td></tr>
<tr><td>10.

</td><td>$(-2,0)$ $(-2,1)$
$\frac{1-0}{-2-(-2)} = \frac{1}{0}$
m = undefined</td><td>There is none</td><td>$x = -2$</td></tr>
</table>